INFO3450/INFO5355/COMM3450
Group Assignment [A5]
Enrolled in [5355]
Project topic: [Fast Cooking App]
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# A5 Group

# Problem Statement

**User:** Busy graduate students who are new to cooking for themselves.

**Need:** They need a way to easily prepare quick, healthy meals that fit their limited time and ingredient availability, while also allowing them to gain confidence in their cooking skills.

**Problem Statement:** Busy graduate students need efficient cooking solutions that cater to their resourcefulness and confidence-building, as they strive to prepare healthy meals using available ingredients without feeling overwhelmed by the complexity of cooking.

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# 1. Report

## 1.1 Usability Test Plan

**Goal:** Evaluate the usability and efficiency of the app by testing major functionalities relevant to the user's needs.

1. Assess the ease of navigation and usability of core features, including login, ingredient input, and recipe generation.
2. Evaluate how well the app supports ingredient substitution and step-by-step cooking.
3. Understand user reactions to saved recipe functionality.
4. Gather user feedback on the overall design, usability, and feature relevance.

**Test Sessions:** Each session will last approximately 20 minutes and include the following components:

1. Introduction and informed consent.
2. Explanation of the think-aloud protocol.
3. Completion of five tasks.
4. Post-session questionnaire and discussion.

**Ethical Considerations:** Participants are informed they are testing the app, not being tested themselves. Confidentiality and participants' privacy are ensured and they can choose to withdraw at any time. Also, avoid influencing participants' actions or responses during the test.

**Session Protocol:**

**1. Introduction and Consent**

Use the following oral script to obtain informed consent.

- **Facilitator:** "Hi, thank you for taking the time to participate in this usability test. Here's the prototype of a recipe app that we created mainly for busy graduate students who are new to cooking for themselves. We focused on helping them easily prepare quick, healthy meals that fit their limited time and ingredient availability, while also allowing them to gain confidence in their cooking skills. I'll be guiding you through the session today. This is part of a project to test a prototype we're working on. The purpose of this session is to understand how intuitive and usable the prototype is for someone like you. There are no right or wrong answers—this is not a test of your abilities but of the design. Your honest feedback is extremely valuable. Before we start, do you have any questions about the session or the process?"
- Obtain verbal or written consent.

**2. Think-Aloud Protocol**

- **Facilitator:** "During this session, please think out loud as you go. Tell me what you're trying to do, what you're thinking, how you feel, and what you are looking for. If anything is unclear or frustrating, let me know. If at any point you get stuck, feel free to continue with your understanding. There are no right or wrong actions. I might also ask you some follow-up questions during or after each task."

**3. Tasks Execution**

**Introduction:**

Imagine you're using the app to cook a meal with ingredients you already have at home. I'll give you a series of tasks to complete. Please follow the instructions and think out loud as you go.

**Task Instructions:**

***Task 1: Login***

- **Scenario:** You want to cook something using the ingredients you have at home. Open the app and log in.
- **Goal:** Test the ease of locating and using the login feature.
- **Follow-up:** Was it easy for you to find and use the login feature?

***Task 2: Custom Recipe Generation***

- **Scenario:** Generate a custom recipe based on available ingredients. Try both the scan and input functions and apply a "one-pan" filter.
- **Goal:** Evaluate the ingredient logging process, the application of filters, and recipe generation functionality.

- **Follow-up:** Did the ingredient scanning and manual input features meet your expectations?

***Task 3: Ingredient Substitution***

- **Scenario:** You want to cook a veggie stir-fry but don't have lettuce. Find a suitable substitution for lettuce using the app and decide if you want to purchase the ingredient or use a substitute.
- **Goal:** Assess the intuitiveness and usability of the substitution feature.
- **Follow-up:** Did the substitution suggestions make sense and feel helpful?

***Task 4: Step-by-Step Cooking Mode***

- **Scenario:** Start the step-by-step mode for the recipe of stir-fried veggies. Follow the instructions to complete the recipe. Pause at Step 2, then resume and move to Step 3.
- **Goal:** Test the clarity of step-by-step mode, timer functionality, and navigation controls.
- **Follow-up:** How helpful were the step-by-step instructions? Did the timer and navigation buttons (pause, previous, next) work as expected?

***Task 5: Save Recipe***

- **Scenario:** Save the recipe for future use and find the saved recipes on the user page.
- **Goal:** Assess the ease of saving and retrieving recipes.
- **Follow-up:** Did you need help finding the saved recipes section?

**4. Post-Session Questionnaire and Discussion**

- What did you like most about the app? Why?
- What did you find most frustrating about the app? Why?
- What would it be if you could change one thing about this app?
- What would make this app more enjoyable or useful for you?
- Do you feel this app addresses the needs of someone like you? Why or why not?
- Would you consider using this app regularly? Why or why not?
- How could this app better support your cooking habits and goals?
- Do you have any additional feedback or suggestions that haven't been addressed?
- Would you recommend this app to others? Why or why not?

**Justification of the Plan:** The usability test plan is designed to thoroughly evaluate the app's usability and efficiency while aligning with the project goals and the persona's goals. The following points justify the design of the plan and its details.

- ❖ **Alignment with Persona's Needs:** Tasks are designed to mirror real-life scenarios that align with the persona's goals. Each task represents real-world scenarios relevant to the personal, such as cooking with available ingredients, managing substitutions, and saving recipes for future use. This ensures that the test evaluates whether the app supports the user's goals of gaining confidence in cooking while effectively preparing meals.

- ❖ **Coverage of Key Features:** Tasks comprehensively cover the app's core functionalities, ensuring all major interactions are tested (e.g., scanning, substitution, and generator, step-by-step mode). This ensures all major features of the app are evaluated, highlighting potential usability issues across the test.
- ❖ **Real-Life Simulation:** The tasks are formulated to reflect real-life scenarios, such as cooking a veggie stir-fry and managing ingredients. Providing participants with practical interactions and they are more likely to provide insightful feedback and help identify usability issues they may encounter.
- ❖ **Structured Tasks and Protocol:** The tasks are structured, clear, and concise. This provides enough information to motivate participants without being stuck or lost. Thus, the test will allow us to observe natural user behavior and identify potential improved features. Additionally, the protocol includes clear oral instructions for facilitating and ensuring consistency across sessions, from obtaining consent to finishing post-session Q&A.
- ❖ **Think-Aloud Protocol Insights:** Encouraging participants to articulate their thoughts and ensuring grabbing insights from their experience and challenges. This method provides valuable feedback from learning users' decision-making processes in each task. This allows us to identify underlying design flaws and improve the user experience.

By aligning with users' goals, covering key features, simulating real-world scenarios, and promoting insightful feedback through tasks and think-aloud protocol, this usability test plan ensures a comprehensive evaluation of the product. The structured approach provides flexibility during the test and aligns with the project's goals, getting feedback from participants for impactful design improvements.

## 1.2 Design improvements & final design

1. **Interactive Search Box on the *Recipe Generator* Page**
   - ***Before***: The yellow search box on the "Recipe Generator" page was not clickable, violating the common expectation that search boxes should be interactive and lead users to a search results page. During usability testing, users repeatedly clicked on the search box, only to find it unresponsive, leading to confusion and frustration.
   - ***After***: The search box is now fully interactive, directing users to the generated recipe page upon clicking. This enhancement creates a more logical and seamless user flow, allowing users to immediately tap the 'AI Generate' icon after entering their quick filter keywords. This improvement aligns with user expectations and significantly enhances usability.

2. **The "Done" Button on the Quick Filter**
   - ***Before***: Previously, after users selected their quick filter, they had to click the "Close" button to apply the filter on the Recipe Generator page. However, this contradicts common design logic, as a "Close" button, especially in the top-right corner, is

typically associated with canceling an action. This design flaw caused confusion and unnecessary delays, as users were unsure how to proceed.

- ***After***: By adding a "Done" button, users can now easily apply their selected filters by clicking "Done," while the "Close" button remains available to cancel the action. This simple yet intuitive redesign aligns with common user expectations and improves the overall usability and efficiency of the page.

3. **Clickable Buttons with Words**
   - ***Before***: Previously, when users attempted to navigate to the next page by clicking buttons, the functionality only worked if they clicked directly on the button itself, not the accompanying words. During usability tests, a participant consistently clicked on the words, assuming they were also clickable, which caused confusion and disrupted the user experience.
   - ***After***: By making both the buttons and their accompanying words clickable, we significantly enhance the user experience. The larger clickable area aligns with user expectations and common design practices, providing a more intuitive and seamless interaction that better supports the natural user flow.

## 1.3 Team Contribution

- ly445: merged heuristic evaluation and usability test Excel forms and came up with suggested solutions. Worked on Usability Test Plan, Figma Prototype, and Design improvements & final design. Developed the usability test plan, including the session protocol, and task instructions. Enhanced the Figma Prototype by implementing design improvements based on feedback from heuristic evaluations and usability tests. Contributed to finalizing the Design Improvements & Final design section by articulating the justification of "before" and "after" states of key changes.
- xc535: merged heuristic evaluation and usability test Excel forms and came up with suggested solutions. Worked on Usability Test Plan. Created the usability test plan, with a focus on aligning tasks and scenarios with users' goals and persona's needs, and justification of the plan. Assisted in reviewing session protocol and task instructions to ensure alignment with usability principles and persona expectations.
- xl992: merged heuristic evaluation and usability test Excel forms and came up with suggested solutions. Worked on Figma Prototype and Design improvements & final design. Made significant changes to the Figma Prototype, implementing high-fidelity design improvements where identified issues. Documented impactful changes, evaluated solutions, and ensured the final portfolio-ready prototype.

## 2. Heuristic Evaluation

https://docs.google.com/spreadsheets/d/1rgTwKpDR8ctVHipEBb5M8HCIo4eg4wH2C8upPtff9R4/edit?usp=sharing

## 3. Usability Test Data Analysis

https://docs.google.com/spreadsheets/d/1WjTOgfqAbdCp0mTp6NfL8OMNoVzTOK2gB1h8vGMnvlM/edit?usp=sharing

## 4. Figma prototype

Before:https://www.figma.com/design/hCgZjyPEqWfoazbzhld1pF/prototype?node-id=0-1&t=QXDWXP0GFlo56MUo-1

After:https://www.figma.com/design/hCgZjyPEqWfoazbzhld1pF/prototype?node-id=374-292&t=RfcdKNvrrE8AbhKN-1

## 5. Showcase Movie

fast cooking app video.mp4